



DATA STRUCTURES II

► *heapify demo*

Lecture slides by Kevin Wayne

<http://www.cs.princeton.edu/~wayne/kleinberg-tardos>

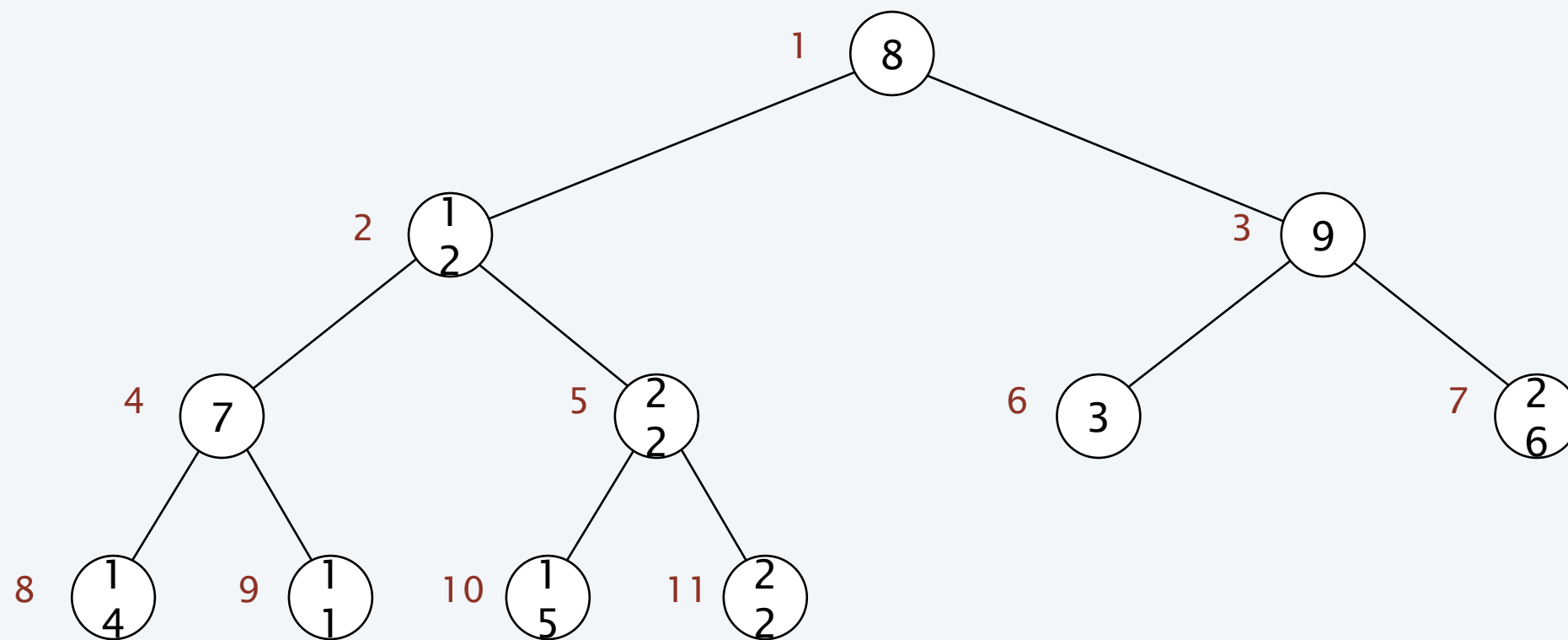
Abdur Rashid Tushar
Lecturer, CSE, BUET

Heapify demo

Heapify. For each element in reverse-array order, sink it down.

we assume array entries are indexed 1 to n

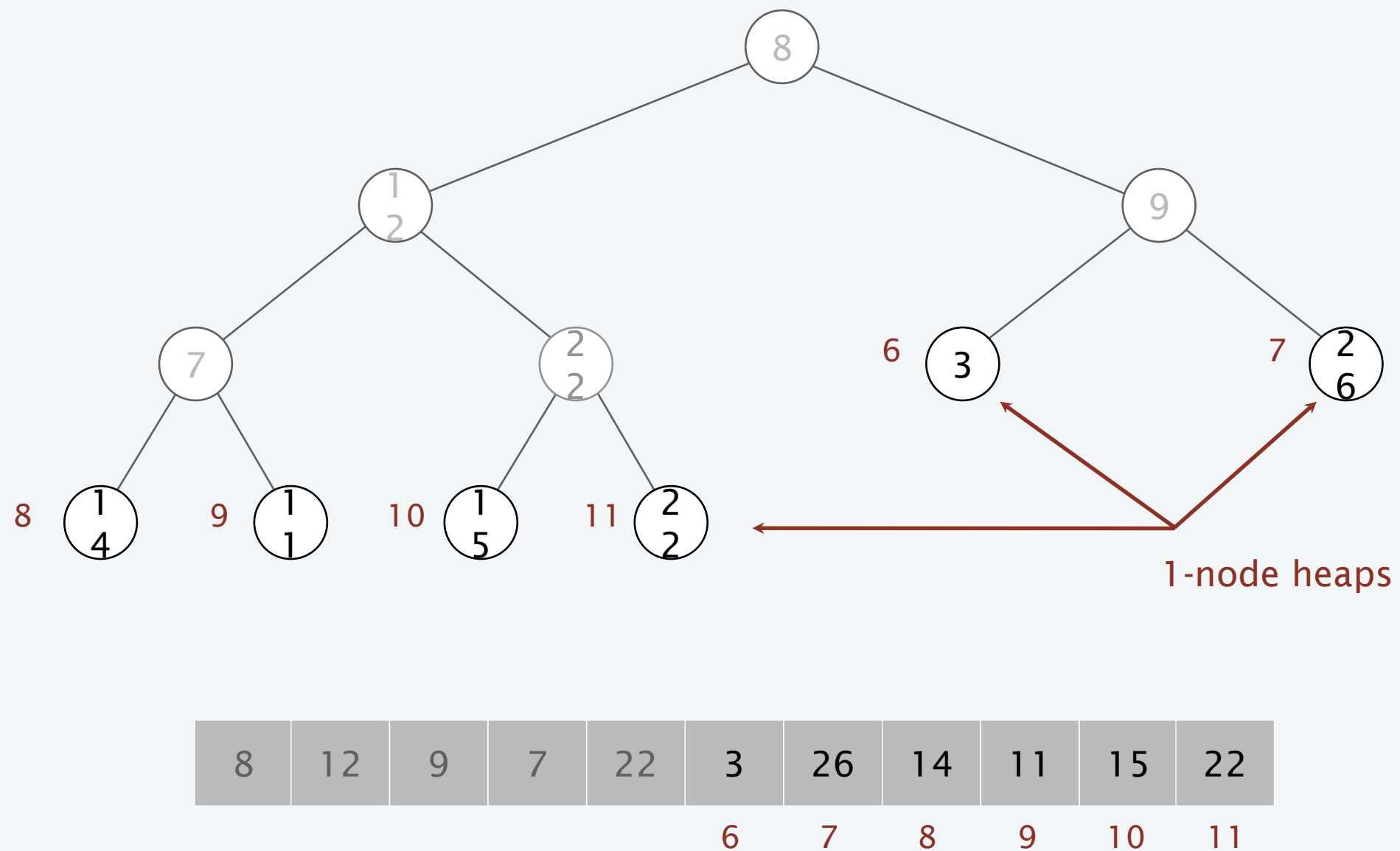
array in arbitrary order



8	12	9	7	22	3	26	14	11	15	22
1	2	3	4	5	6	7	8	9	10	11

Heapify demo

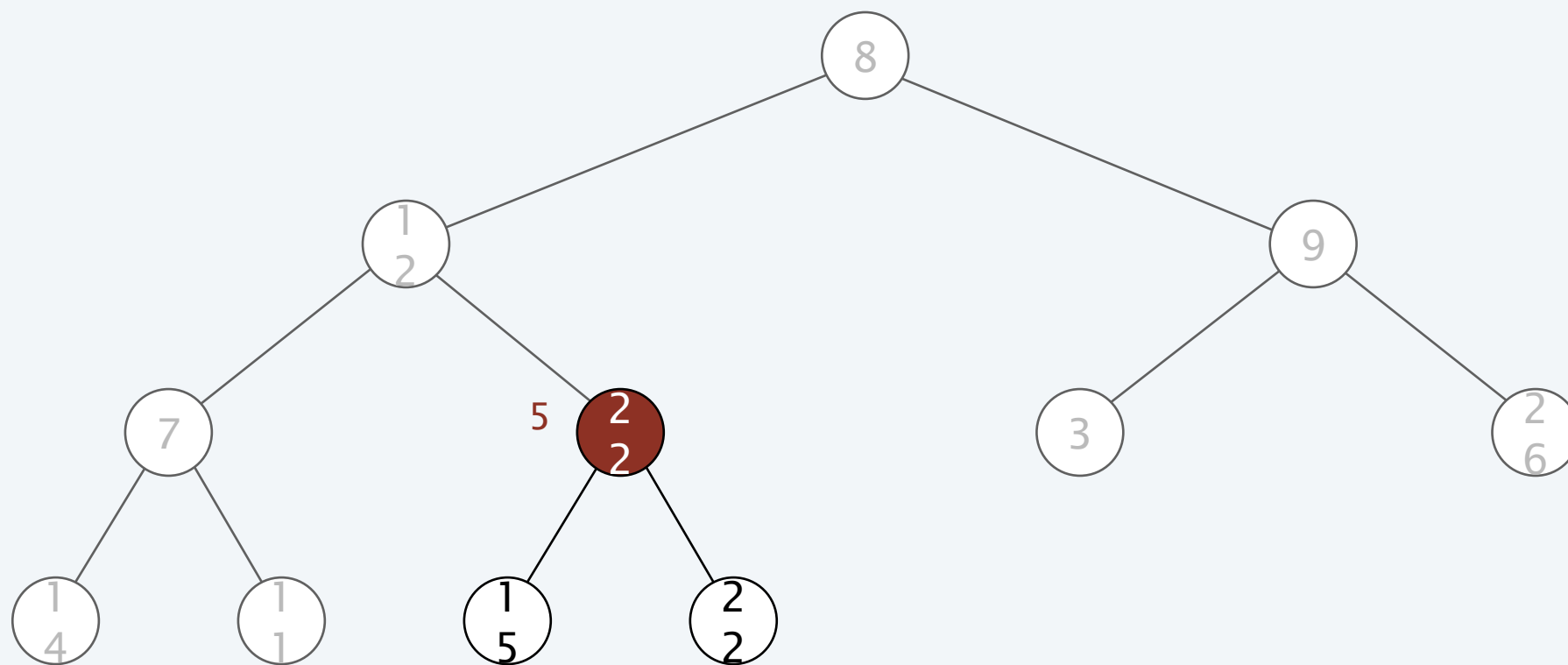
Heapify. For each element in reverse-array order, sink it down.



Heapify demo

Heapify. For each element in reverse-array order, sink it down.

sink 5



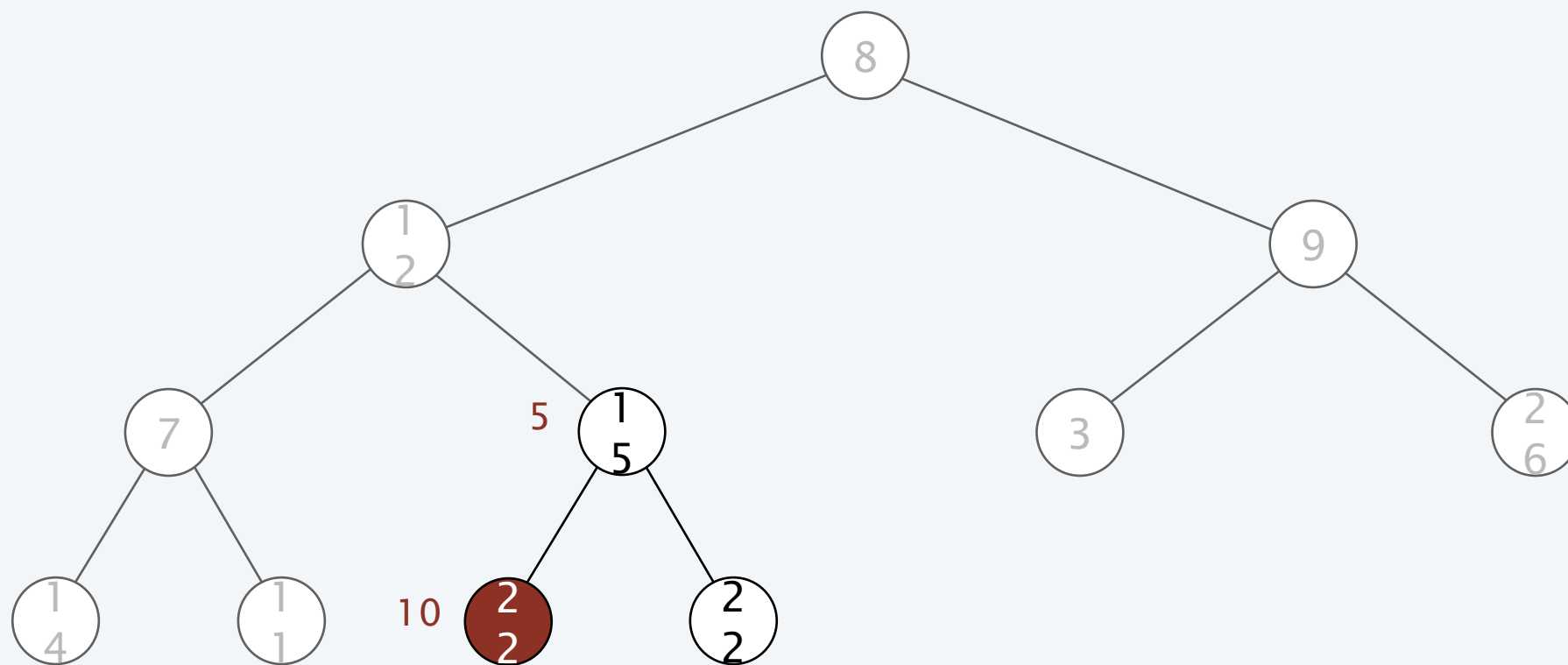
8	12	9	7	22	3	26	14	11	15	22
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5

Heapify demo

Heapify. For each element in reverse-array order, sink it down.

sink 5

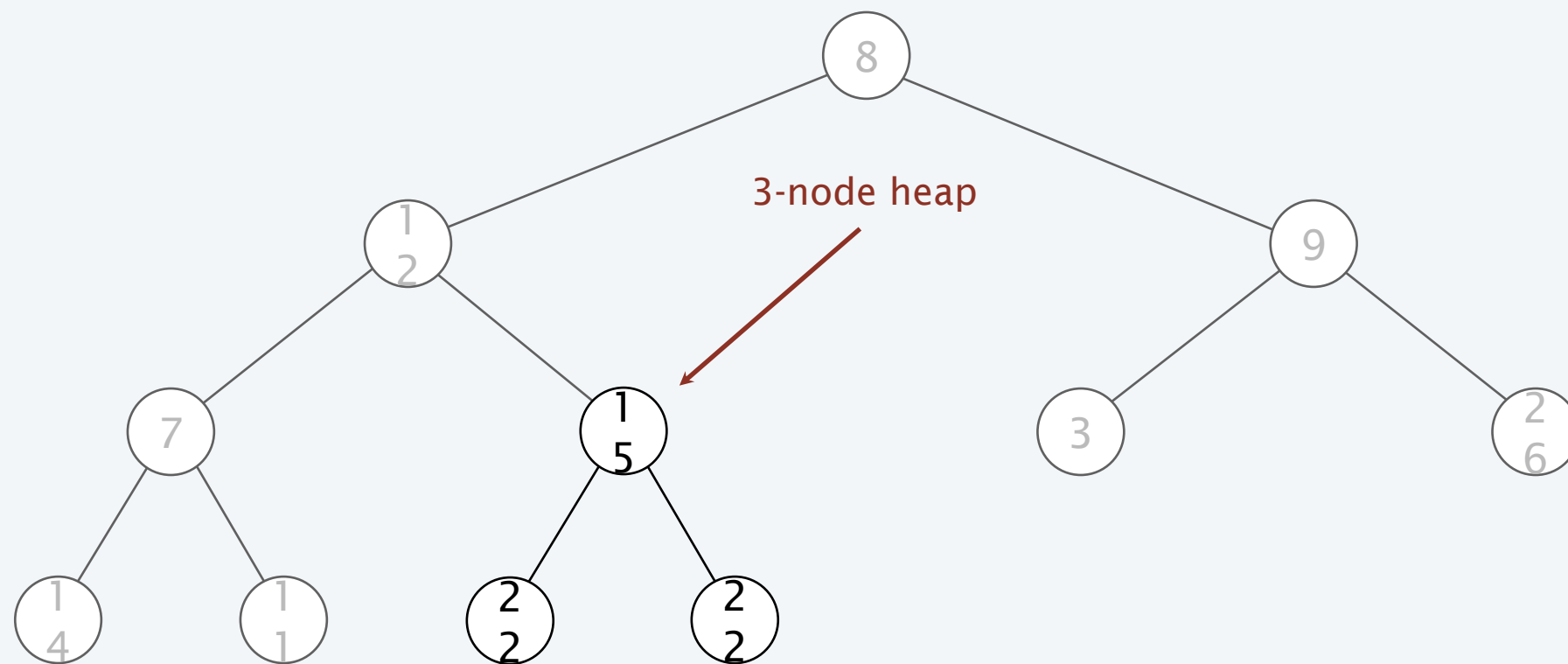


8	12	9	7	15	3	26	14	11	22	22
				5				10		

Heapify demo

Heapify. For each element in reverse-array order, sink it down.

sink 5

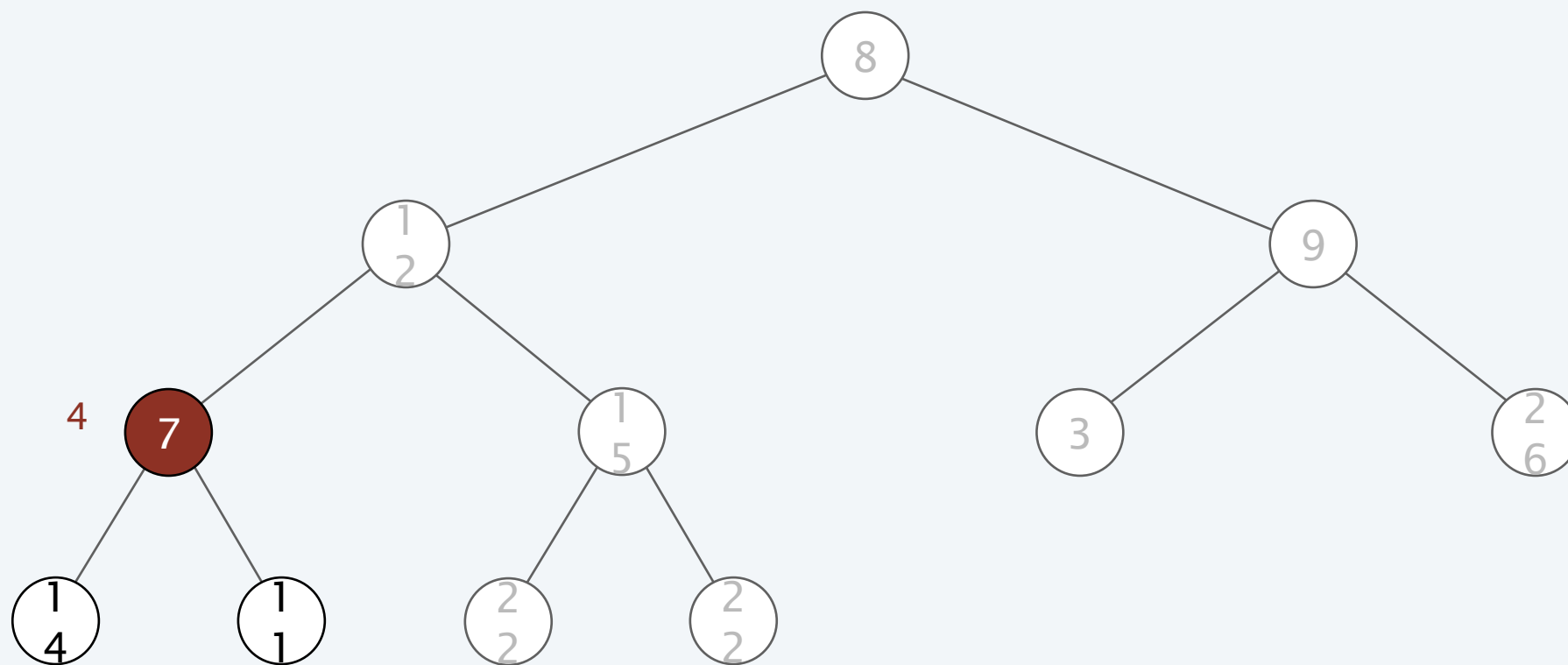


8	12	9	7	15	3	26	14	11	22	22
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Heapify demo

Heapify. For each element in reverse-array order, sink it down.

sink 4



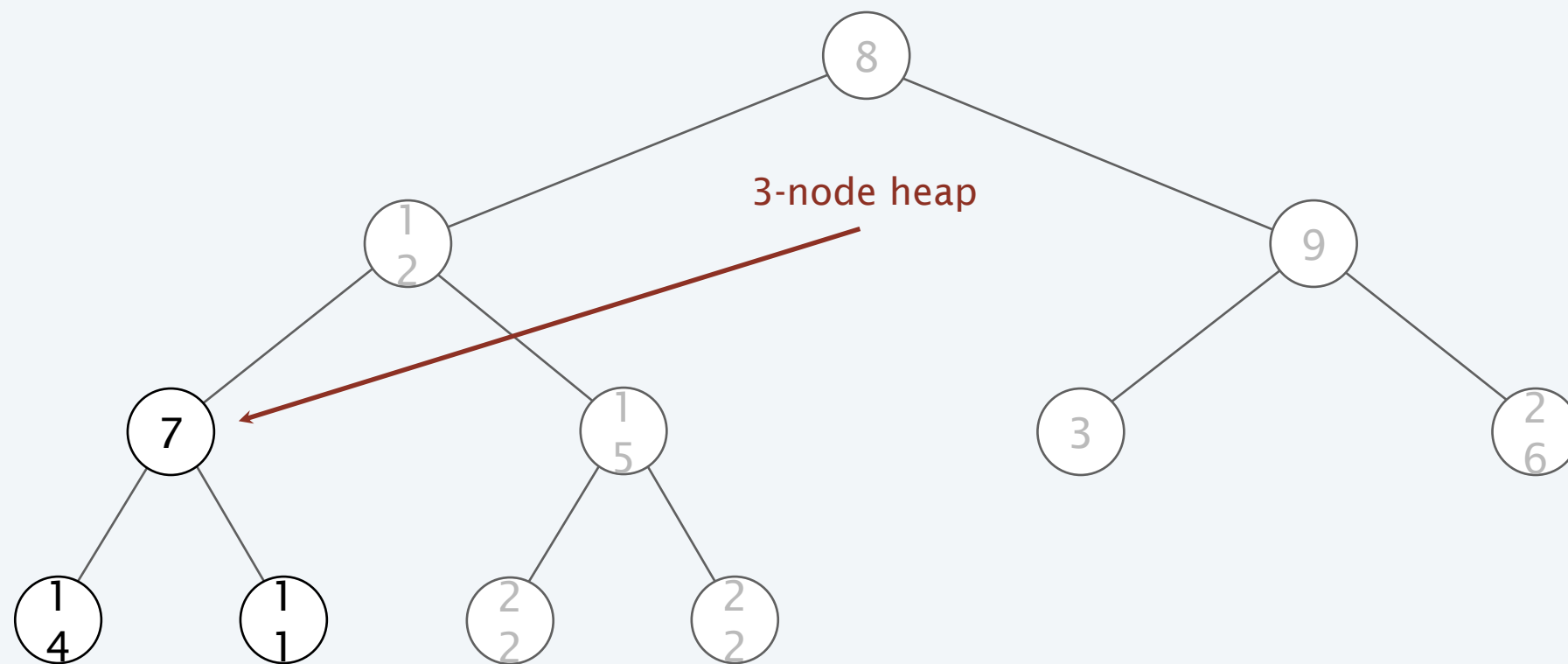
8	12	9	7	15	3	26	14	11	22	22
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4

Heapify demo

Heapify. For each element in reverse-array order, sink it down.

sink 4

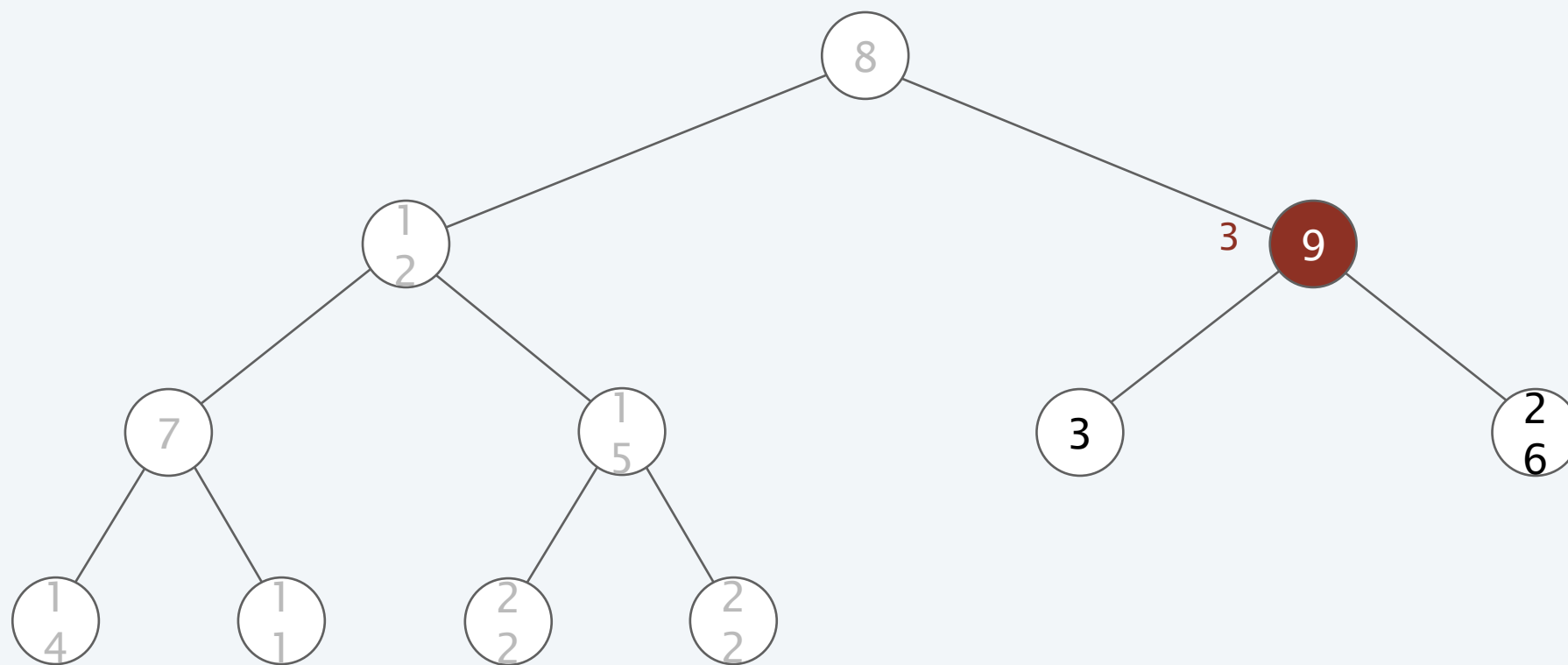


8	12	9	7	15	3	26	14	11	22	22
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Heapify demo

Heapify. For each element in reverse-array order, sink it down.

sink 3

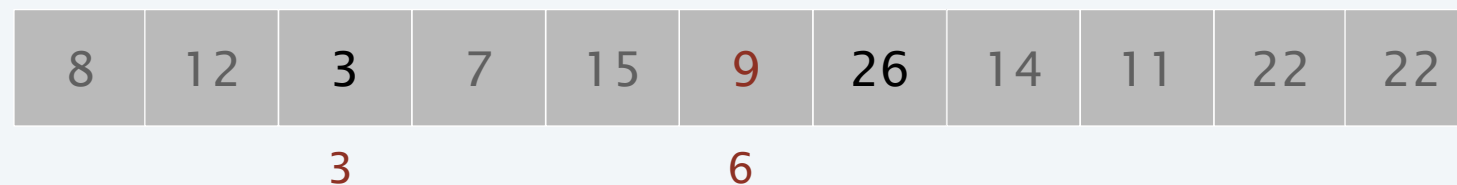
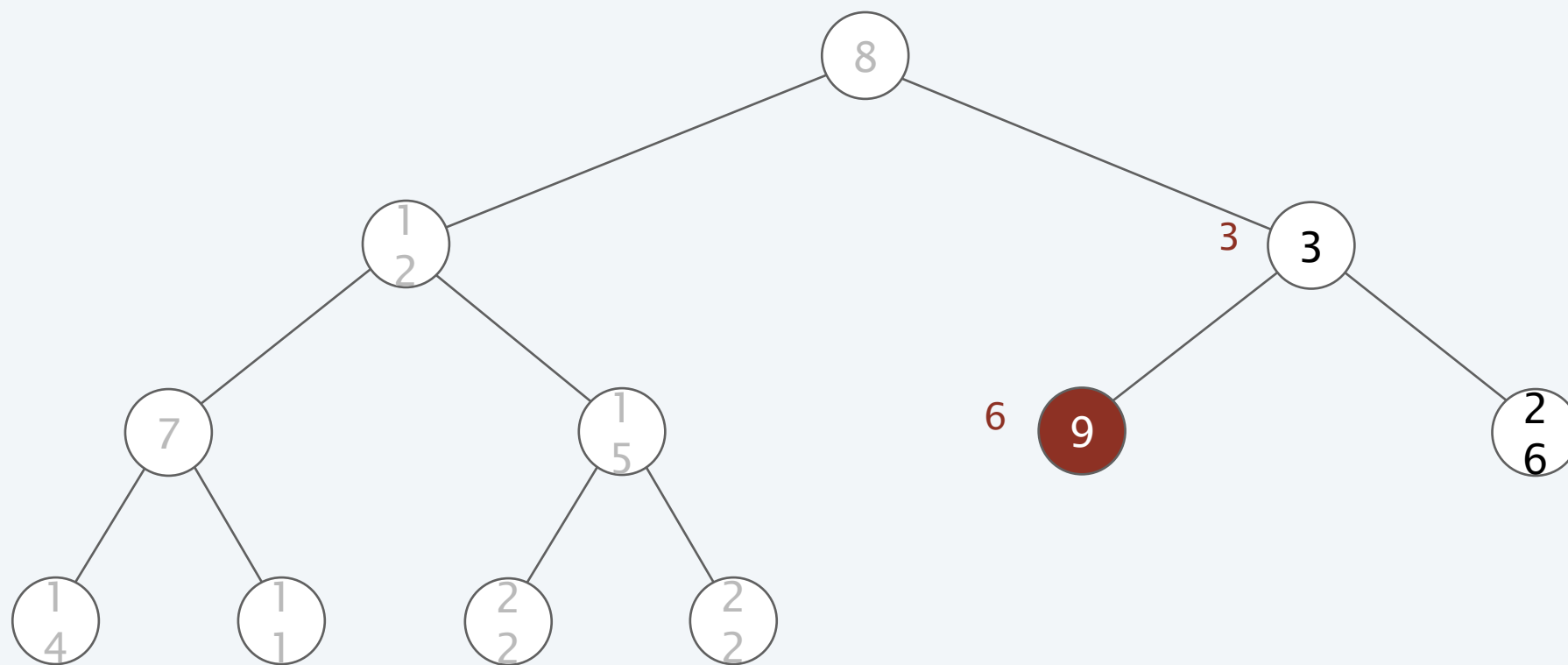


3

Heapify demo

Heapify. For each element in reverse-array order, sink it down.

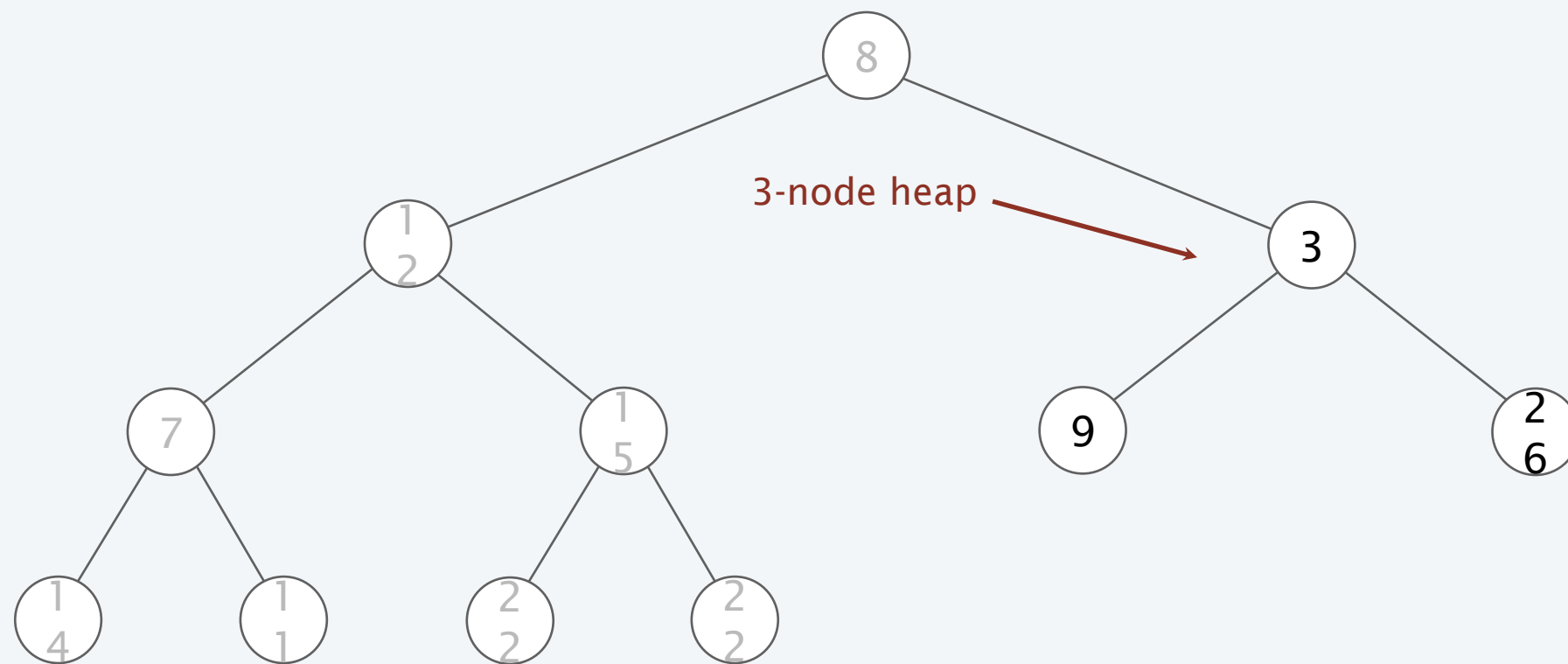
sink 3



Heapify demo

Heapify. For each element in reverse-array order, sink it down.

sink 3

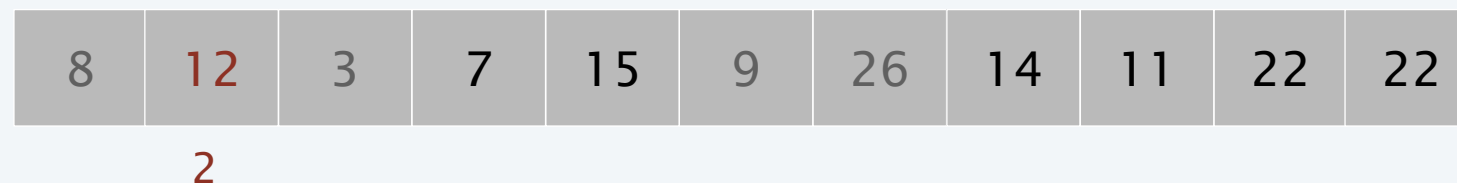
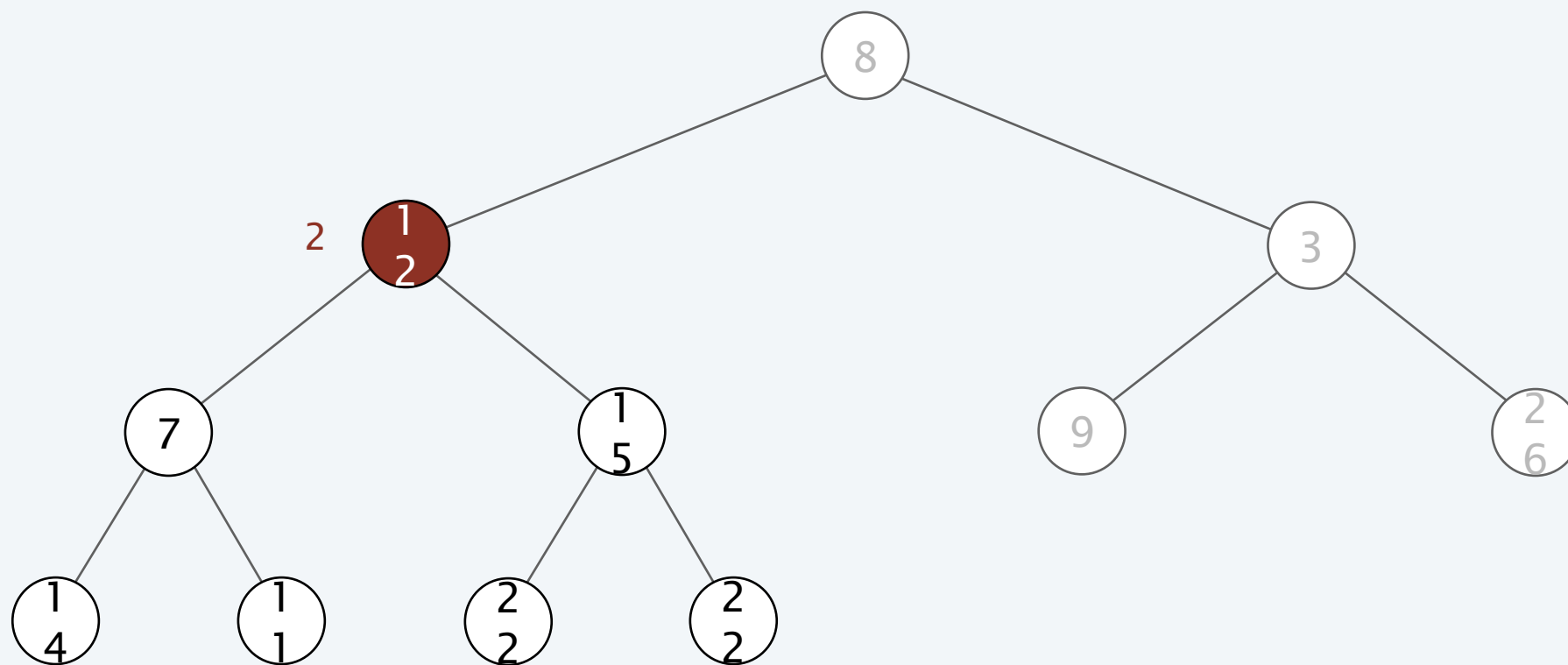


8	12	3	7	15	26	26	14	11	22	22
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Heapify demo

Heapify. For each element in reverse-array order, sink it down.

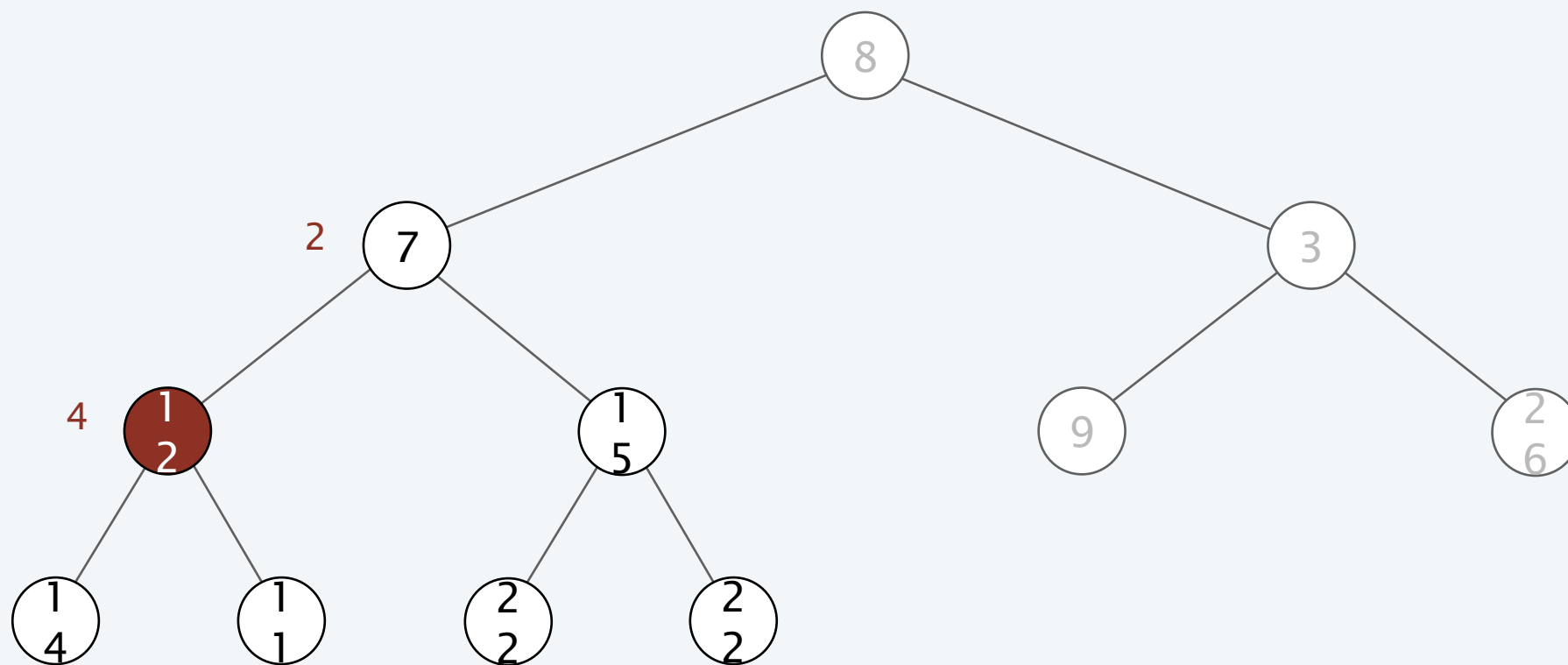
sink 2



Heapify demo

Heapify. For each element in reverse-array order, sink it down.

sink 2

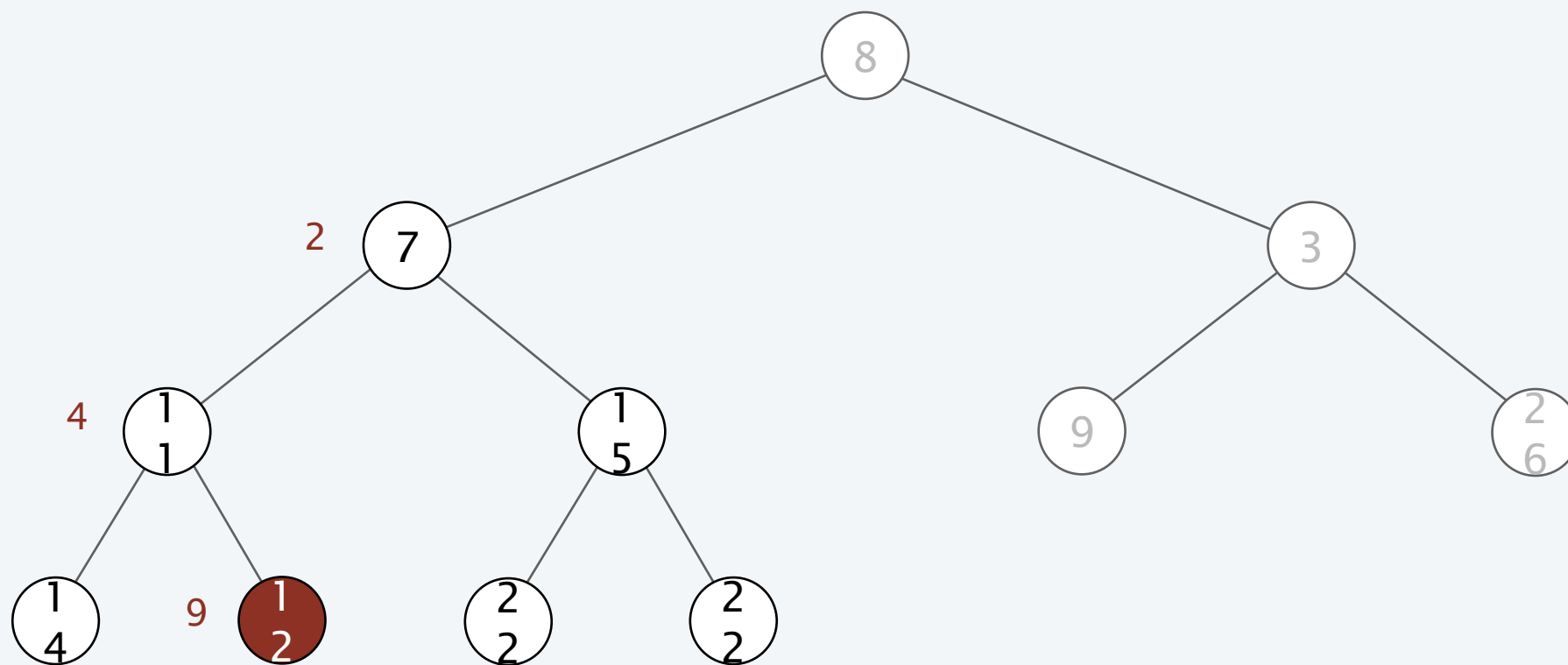


8	7	3	12	15	9	26	14	11	22	22
	2		4							

Heapify demo

Heapify. For each element in reverse-array order, sink it down.

sink 2

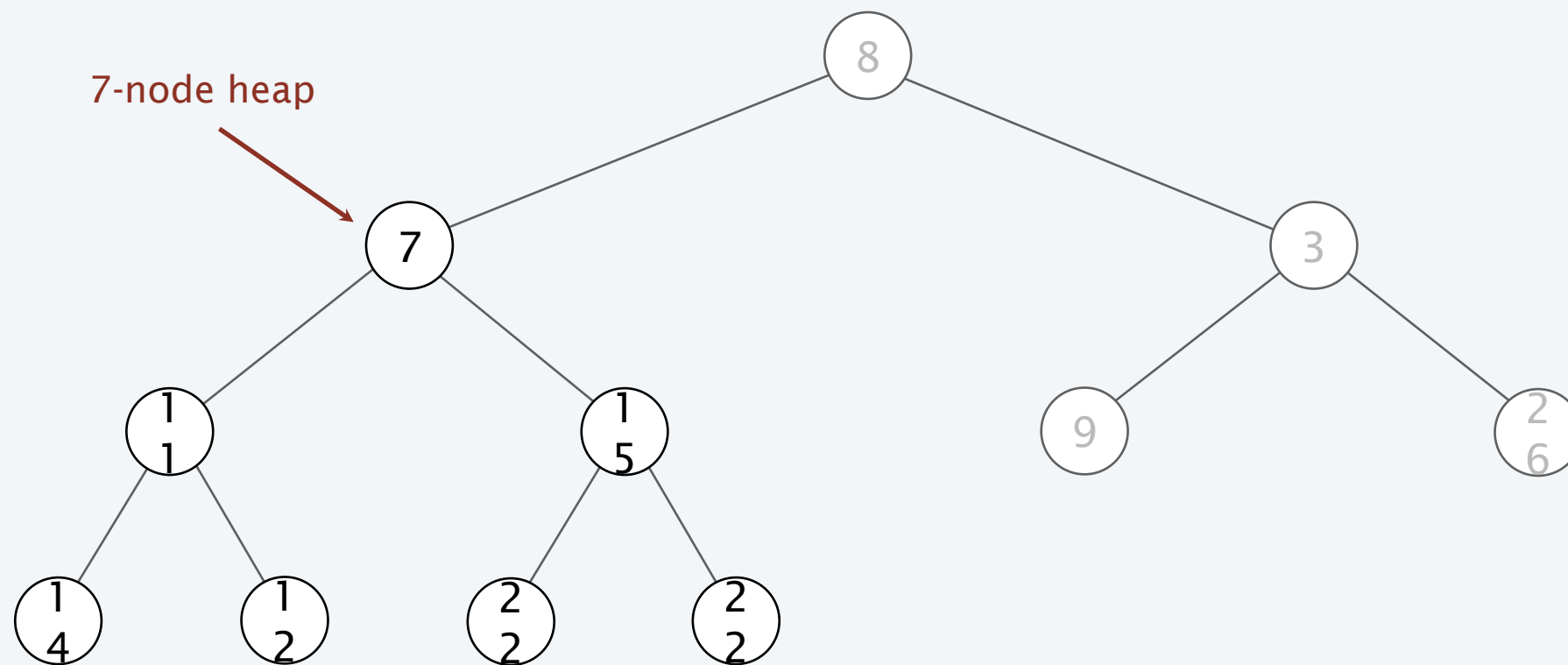


8	7	3	11	15	9	26	14	12	22	22
	2		4					9		

Heapify demo

Heapify. For each element in reverse-array order, sink it down.

sink 2

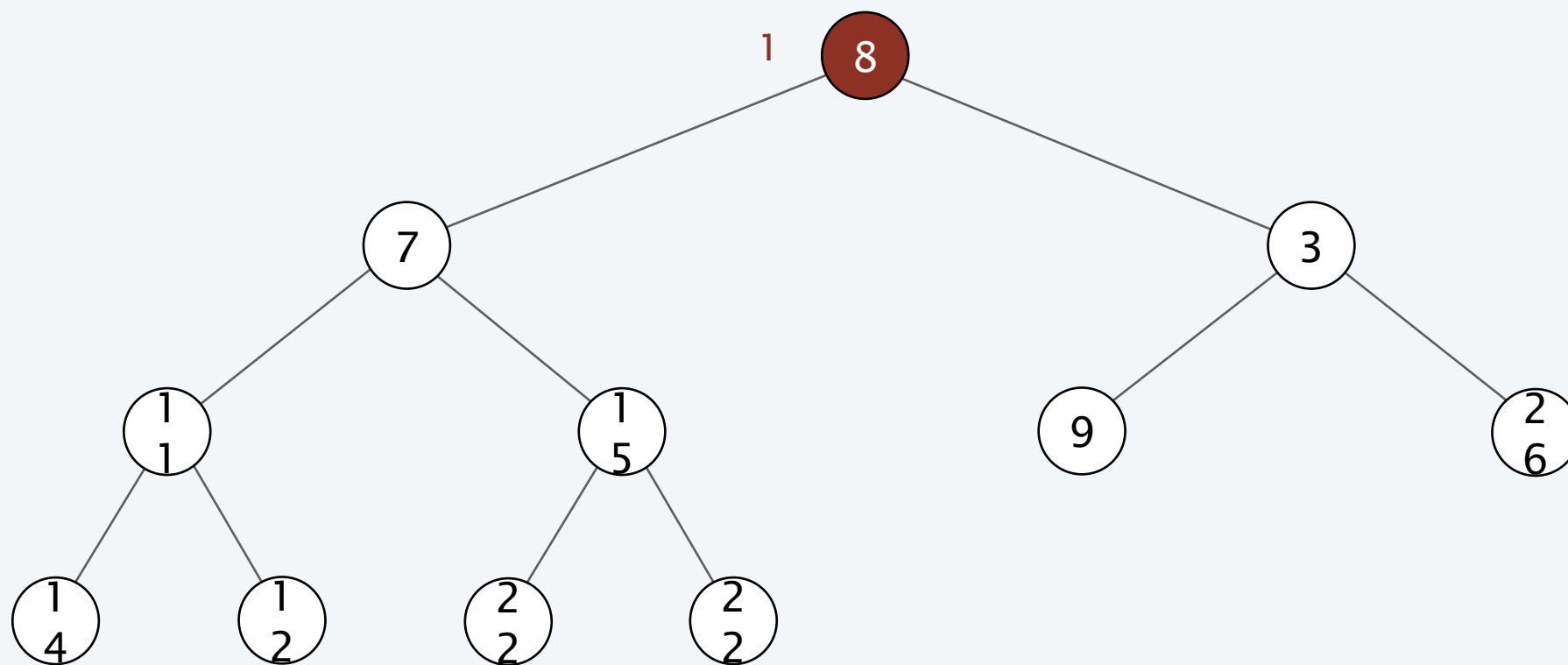


8	7	3	11	15	9	26	14	12	22	22
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Heapify demo

Heapify. For each element in reverse-array order, sink it down.

sink 1

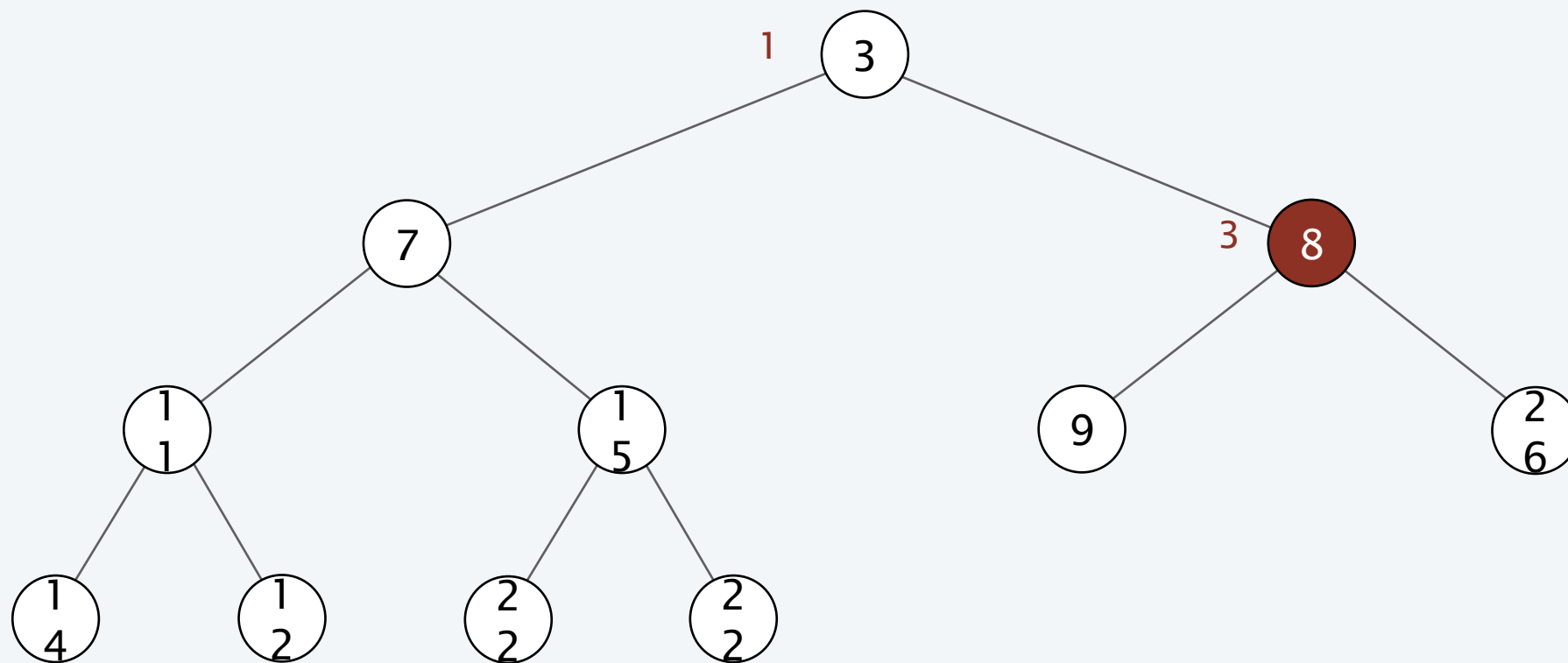


8	7	3	11	15	9	26	14	12	22	22
1										

Heapify demo

Heapify. For each element in reverse-array order, sink it down.

sink 1

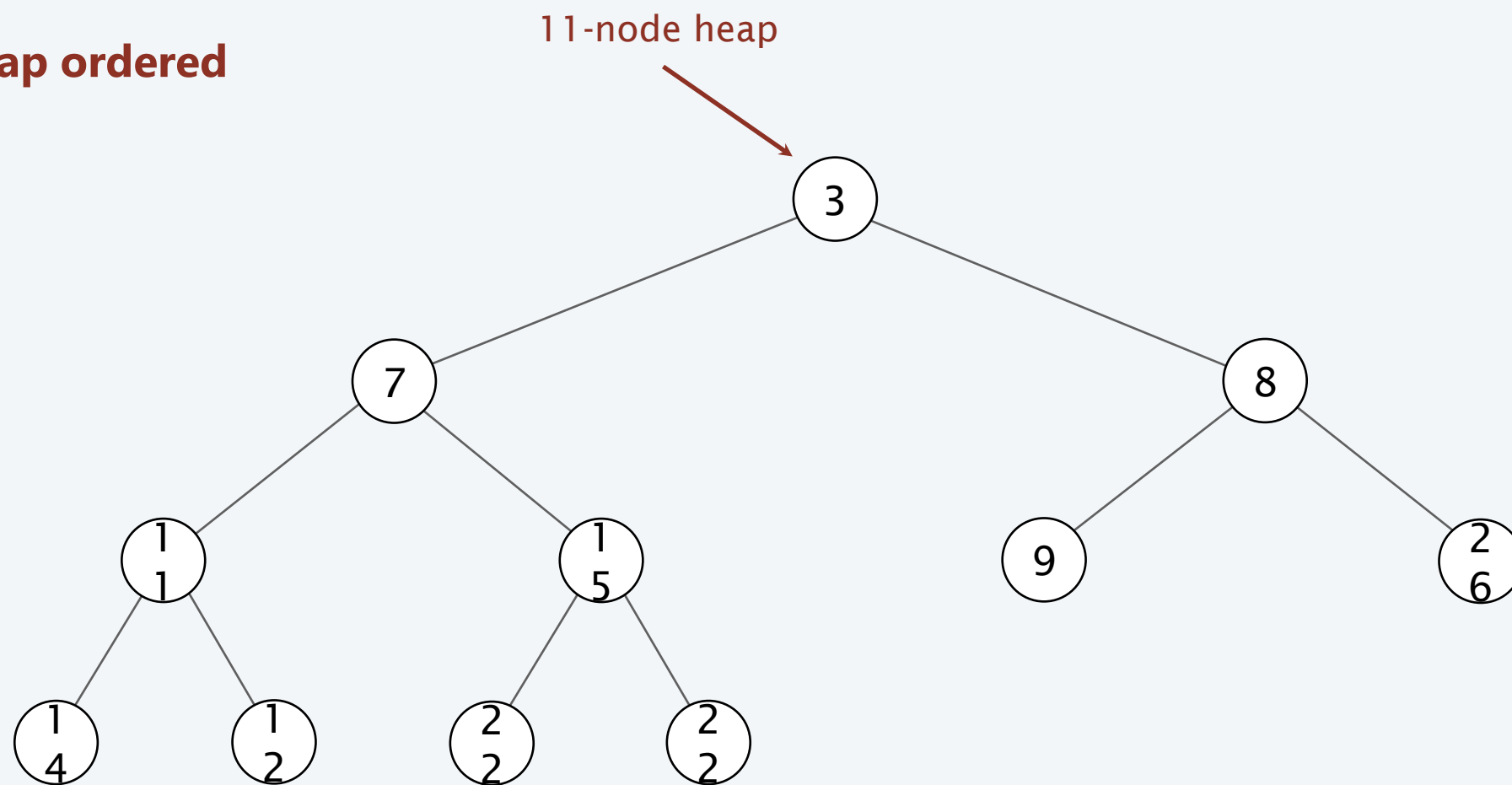


3	7	8	11	15	9	26	14	12	22	22
1		3								

Heapify demo

Heapify. For each element in reverse-array order, sink it down.

array is heap ordered



3	7	8	11	15	9	26	14	12	22	22
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